

MICHIGAN CITY ENTRANCE LIGHT, IN, USA

WMO: 997339

Lat: 41.7290N

Lon: 86.9117W

Elev: 646

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.8	9.5	N/A	N/A	N/A	N/A	N/A	N/A	41.7	21.8	39.1	23.2	16.1	190	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	87.4	N/A	85.0	N/A	82.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 36.9	32.8	28.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	28.8	26.2	37.6	48.9	59.1	68.2	73.8	72.0	65.5	54.7	42.8	33.9	(6)
(7)		DBStd	18.33	9.83	10.09	10.21	8.57	9.29	7.15	5.98	4.88	6.58	8.13	9.49	9.08	(7)
(8)		HDD50	2679	660	666	413	124	20	0	0	0	0	38	254	504	(8)
(9)		HDD65	5885	1123	1085	852	490	235	49	6	5	71	339	667	963	(9)
(10)		CDD50	3079	2	0	28	90	302	547	738	682	464	183	37	6	(10)
(11)		CDD65	809	0	0	2	5	51	146	279	222	85	19	0	0	(11)
(12)		CDH74	4638	0	0	11	39	359	839	1841	1052	401	96	0	0	(12)
(13)		CDH80	1096	0	0	0	3	97	190	501	207	82	16	0	0	(13)
(14)	Wind	WSAvg	13.6	16.4	15.0	14.0	14.5	12.1	10.2	10.5	10.2	12.1	15.4	16.5	15.6	(14)
(15)	Precipitation	PrecAvg	39.3	2.6	2.1	2.5	3.5	3.7	4.3	4.1	3.4	3.5	3.1	2.6	(15)	
(16)		PrecMax	56.8	5.7	4.2	5.2	6.8	6.9	10.0	7.2	9.8	13.6	8.9	7.5	5.0	(16)
(17)		PrecMin	29.9	0.8	0.8	0.8	1.1	0.6	1.5	1.6	0.9	0.7	0.8	0.6	0.8	(17)
(18)		PrecStd	6.1	1.3	1.0	1.1	1.5	1.9	2.1	1.5	2.3	2.6	2.0	1.5	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	58.3	76.2	79.4	87.7	88.4	91.3	88.7	86.8	82.8	69.2	61.5	(19)
(20)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	52.5	48.6	68.5	73.5	82.7	85.1	87.9	85.4	82.1	76.6	64.5	56.5	(21)
(22)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	47.3	42.4	61.1	68.8	77.9	81.9	85.4	82.3	78.6	71.9	60.3	50.8	(23)
(24)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	42.2	37.9	53.6	63.9	73.4	79.0	82.7	79.5	75.2	67.4	56.5	46.5	(25)
(26)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)
(28)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range		MDBR	10.6	11.2	12.2	14.2	14.3	12.5	11.1	10.7	11.6	12.0	11.6	9.7	(35)
(36)		5% DB	MCDBR	16.2	17.1	22.4	22.6	20.1	16.4	14.6	14.5	15.0	17.1	16.3	14.5	(36)
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.330	0.348	0.402	0.432	0.436	0.450	0.447	0.421	0.359	0.324	0.305	(40)	
(41)		taud	2.431	2.348	2.373	2.251	2.227	2.265	2.225	2.229	2.264	2.440	2.492	2.489	(41)	
(42)		Ebn at Noon	267	277	285	275	269	267	262	259	257	263	259	259	(42)	
(43)		Edh at Noon	25	31	34	42	44	43	44	42	38	29	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	479	743	1131	1475	1682	1928	1946	1687	1388	905	562	401	(44)	
(45)		RadStd	48	97	89	124	149	122	96	99	116	96	72	43	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page